4-28-26 (WEEK 10)

Shun/修海 (@shun4midx)

FUNCTIONS OF CONTINUOUS RANDOM VARIABLES

Let X be a continuous random variable with PDF f_X. $X: S \to \mathbb{R}$

Let $Y = g(X)$ for some $g: \mathbb{R} \to \mathbb{R}$

Then Y is also a random variable $Y = g \circ X: S \to \mathbb{R}$

What can we say about its distribution?

Process

① Find the CDF $F_Y(y) := \mathbb{P}(Y \le y)$

② Find the PDF $f_Y(y) = F_Y'(y)$

EXAMPLE

Suppose $Y = X^2$.

Then, $F_Y(y) = \mathbb{P}(Y \le y) = \mathbb{P}(X^2 \le y)$ (=0 if $y<0$, so WLOG assume $y \ge 0$ from now on)

$= \mathbb{P}(-\sqrt{y} \le X \le \sqrt{y})$

$= \int_{-\sqrt{y}}^{\sqrt{y}} f_X(x)\,dx$

THEOREM

Let $Y = g(X)$, where g: diffable and strictly increasing ($g^{-1}(\cdot)$: well-def)

Then, if X has PDF f_X, the PDF of Y is:

$$f_Y(y) = \begin{cases} 0, & y \notin g(\mathbb{R}) \\ f_X(g^{-1}(y)) \cdot \frac{d}{dy}(g^{-1}(y)), & \text{o/w} \end{cases}$$

(Rmk: $\frac{d}{dy}(g^{-1}(y)) = \frac{1}{g'(g^{-1}(y))}$)

Proof

Recall, $\mathbb{P}(Y \le y) = \mathbb{P}(g(X) \le y)$

$$= \begin{cases} 0 & , y \le \inf g(\mathbb{R}) \\ \mathbb{P}(X \le g^{-1}(Y)) = F_X(g^{-1}(Y)), & y \in g(\mathbb{R}) \\ 1 & , y \ge \sup g(\mathbb{R}) \end{cases}$$

where $F_X(x) = \mathbb{P}(X \le x)$ is the CDF of X

To find the PDF of Y, we diff:

$$f_Y(y) = \frac{d}{dy}(\mathbb{P}(Y \le y)) = \begin{cases} 0, & y \notin g(\mathbb{R}) \\ f_X(g^{-1}(y)) \frac{d}{dy}(g^{-1}(y)), & y \in g(\mathbb{R}) \end{cases}$$

JOINTLY DISTRIBUTED RANDOM VARIABLES

EXAMPLE

$X \sim Bin(n,p)$, $Y \sim Geom(p)$, $0 \le p \le 1$, $n \in \mathbb{N}$.

What is $\mathbb{P}(X=k, Y=n)$? What is $\mathbb{E}[X+Y]$?

Ans

Linearity of expectation $\Rightarrow \mathbb{E}[X+Y] = \mathbb{E}[X] + \mathbb{E}[Y] = np + \frac{1}{p}$

However, $\mathbb{P}(X=k, Y=n) \overset{?}{=} \mathbb{P}(X=k)\mathbb{P}(Y=n)$

Say we have a p-biased coin ($\mathbb{P}(\text{Heads}) = p$)

X = # heads in n tosses

Y = # tosses until first heads

If we toss n times, count heads (X), then start over and count tosses to first heads (Y), then they're indep $\Rightarrow$ ineq true

But if we use the SAME tosses to measure X and Y, then they're dependent, so ineq false

$\therefore$ We don't know.

However, we do know:

$$\mathbb{P}(X=k, Y=n) = \mathbb{P}(\{X=k\} \cup \{Y=n\}) = \mathbb{P}(X=k)\,\mathbb{P}(Y=n \mid X=k)$$
$$= \mathbb{P}(Y=n)\,\mathbb{P}(X=k \mid Y=n)$$

DEFINITION

Given random variables X, Y in a probability space $(S, \mathbb{P})$, we define the joint cummulative distribution function $F_{X,Y}(a,b) := \mathbb{P}(\{X \le a\} \cap \{Y \le b\}) = \mathbb{P}(X \le a, Y \le b)$

DEFINITION

Given variables X, Y in a probability space $(S, \mathbb{P})$ with joint CDF $F_{X,Y}$, the marginal distributions are $F_X(x) = \mathbb{P}(X \le x) = \mathbb{P}(X \le x, Y \le \infty) = F_{X,Y}(x, \infty) = \lim_{y \to \infty} F_{X,Y}(x,y)$

Similarly, $F_Y(y) = \lim_{x \to \infty} F_{X,Y}(x,y)$

DEFINITION

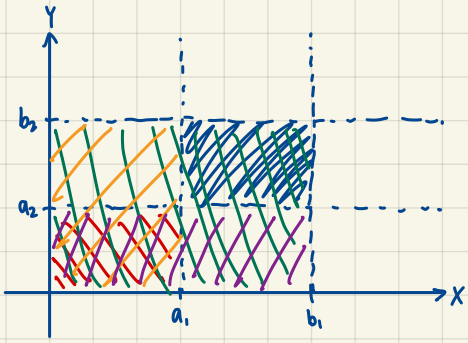
Given discrete random variables X, Y on a probability space $(S, \mathbb{P})$, we define the joint probability mass function $P_{X,Y}(x,y) = \mathbb{P}(X=x, Y=y)$

Note we have the marginals: $P_X(x) = \sum_y P_{X,Y}(x,y)$, $P_Y(y) = \sum_x P_{X,Y}(x,y)$

LEMMA

For any variables X, Y, $\mathbb{P}(\{a_1 \le X \le b_1\} \cap \{a_2 \le Y \le b_2\}) = F_{X,Y}(a_1, a_2) + F_{X,Y}(b_1, b_2) - F_{X,Y}(a_1, b_2) - F_{X,Y}(b_1, a_2)$

Proof



EXAMPLE

In a country:

- 15% of families have 0 children
- 30% of families have 1 child
- 35% of families have 2 children
- 20% of families have 3 children

Each child is equally likely a boy or a girl.

Let X = # boys in a unif rand family
Y = # girls in a unif rand family

What's the distribution of X, Y?
What's $\mathbb{E}[X]$?

Solution

$P_{X,Y}(i,j) = \mathbb{P}(X=i, Y=j)$

Notice, by def of cond prob, $\mathbb{P}(X=i, Y=j, X+Y=i+j) = \cancel{\mathbb{P}(X+Y=i+j \mid X=i, Y=j)}\,\mathbb{P}(X=i, Y=j)$

However, again by def, $\mathbb{P}(X=i, Y=j, X+Y=i+j) = \mathbb{P}(X=i, Y=j \mid X+Y=i+j)\,\mathbb{P}(X+Y=i+j) = \mathbb{P}(\mathrm{Bin}(i+j, \tfrac{1}{2}) = i)\,\mathbb{P}(X+Y=i+j)$

Thus, we can construct the following table of probabilities

X \ Y	0	1	2	3
0	1 (0.15)	$\binom{1}{0}(\frac{1}{2})^0(\frac{1}{2})^1 \cdot 0.30$	$\binom{2}{0}(\frac{1}{2})^0(\frac{1}{2})^2 \cdot 0.35$	$\binom{3}{0}(\frac{1}{2})^0(\frac{1}{2})^3 \cdot 0.20$
1	$\binom{1}{0}(\frac{1}{2})^1(\frac{1}{2})^0 \cdot 0.30$	$\binom{2}{1}(\frac{1}{2})^1(\frac{1}{2})^1 \cdot 0.35$	$\binom{3}{1}(\frac{1}{2})^1(\frac{1}{2})^2 \cdot 0.20$	0
2	$\binom{2}{0}(\frac{1}{2})^2(\frac{1}{2})^0 \cdot 0.35$	$\binom{3}{1}(\frac{1}{2})^2(\frac{1}{2})^1 \cdot 0.20$	0	0
3	$\binom{3}{0}(\frac{1}{2})^3(\frac{1}{2})^0 \cdot 0.20$	0	0	0

$\parallel$

sum the rows

X \ Y	0	1	2	3	$P_X(x)$
0	0.15	0.15	0.0875	0.025	0.4125
1	0.15	0.175	0.075	0	0.4
2	0.0875	0.075	0	0	0.1625
3	0.025	0	0	0	0.025

Hence, $\mathbb{E}(X) = \sum x \mathbb{P}(X=x) = \sum x P_X(x) = 0P_X(0) + P_X(1) + 2P_X(2) + 3P_X(3) = 0.8$

DEFINITION

Given continuous random variables X, Y on a probability space $(S, \mathbb{P})$, we say they have a <u>joint probability density function</u> $f_{X,Y}(x,y)$ if $\forall$ measurable set $C \subseteq \mathbb{R}^2$, $\mathbb{P}((X,Y) \in C) = \iint_C f_{X,Y}(u,v)\,du\,dv$

Also, $F_{X,Y}(x,y) := \mathbb{P}(X \leq x, Y \leq y) = \mathbb{P}((X,Y) \in (-\infty, x] \times (-\infty, y]) = \iint_C f_{X,Y}(u,v)\,du\,dv = \int_{-\infty}^{x}\int_{-\infty}^{y} f_{X,Y}(u,v)\,dv\,du$

Thus, $f_{X,Y}(x,y) = \frac{\partial^2}{\partial u \partial v} F_{X,Y}(u,v)\big|_{u=x, v=y}$

EXAMPLE

Let X, Y have the joint density function

$$f_{X,Y}(x,y) = \begin{cases} e^{-(x+y)}, & x+y \geq 0 \\ 0, & x+y < 0 \end{cases}$$

1. What is $\mathbb{P}(X > Y)$? $\rightsquigarrow$ integrate over the region $X > Y$
2. What is $\mathbb{P}(X \leq t)$, $t \in \mathbb{R}$? $\rightsquigarrow$ integrate over the region $X \leq t$
3. What is the distribution of $\frac{X}{Y}$? $\rightsquigarrow$ Find CDF, then diff to find PDF